

Household Income & Consumption Expenditure Survey

Project Scope & Objectives · Saudi Arabia 2023 · General Authority for Statistics (GASTAT)

Project Name	HICES 2023 — Household Income & Consumption Analysis
Data Source	General Authority for Statistics (GASTAT) — Saudi Arabia
Tools Used	SQL Server (SSMS) · Power BI Desktop · GitHub
Document Type	Project Requirements & Objectives
Phase	Phase 1 — Project Scoping

1 · Project Background

This project delivers a comprehensive data analytics pipeline built on the 2023 Household Income and Consumption Expenditure Survey (HICES) published by the General Authority for Statistics (GASTAT) of Saudi Arabia. HICES is one of the most thorough economic datasets covering Saudi households — capturing disposable income, consumer expenditure, and their distribution across demographic and geographic variables. The project covers the full analytics lifecycle: raw-data cleaning and ingestion into SQL Server, analytical query writing, interactive Power BI dashboard development, and publication on GitHub.

2 · Stakeholder Request

“We need a clear picture of household income and expenditure from the 2023 survey. The analysis must address gaps across nationalities, regions, and education levels, and identify the top spending categories. The final deliverable is an interactive dashboard that can be shared with senior management.”

Three analytical pillars were derived from this request:

- Demographic gap analysis — nationality, age, education, marital status
- Geographic analysis — across all 13 administrative regions
- Expenditure structure analysis — using the COICOP classification

3 · Analytical Objectives

#	Analytical Question	Expected Output
1	What is the income gap between Saudi and non-Saudi households?	Gap ratio + mean & median comparison
2	Which administrative regions have the highest income and expenditure?	Region ranking + household surplus/deficit
3	How does education level affect household income?	Income curve vs. education attainment
4	At what life stage does household income peak?	Economic life-cycle curve
5	What are the largest household consumption expenditure categories?	COICOP ranking + % share of total

6	Do households save or spend beyond their income?	Approximate savings rate by region & demographic
---	--	--

4 • Technical Objectives

- Clean raw Excel data and export into 7 structured CSV files
- Build a relational database in SQL Server and design appropriate schema
- Write analytical SQL queries answering all six analytical questions above
- Export query results as CSV and connect them to Power BI
- Build a 4-page interactive Power BI dashboard covering all analytical pillars
- Fully document and publish the project on GitHub

5 • Project Deliverables

Deliverable	Tool	Phase
Cleaned CSV files (×7)	Excel	Phase 2
Relational Database — HICES_2023	SQL Server / SSMS	Phase 3
Analytical SQL Queries	SSMS	Phase 3
Interactive Dashboard (4 pages)	Power BI Desktop	Phase 4
Documented GitHub Repository	Git / GitHub	Phase 5

6 • Project Workflow

#	Phase	Key Tasks	Status
1	Project Scoping	Write requirements document (this document)	Complete
2	Data Preparation	Clean Excel source + export 7 CSV files	In Progress
3	Database Build	Create DB + tables + import data + write queries	Not Started
4	Power BI Dashboard	Connect data + DAX measures + build dashboard	Not Started
5	GitHub Publication	Document project + upload files + write README	Not Started

7 • Success Criteria (KPIs)

- All six analytical questions fully covered in the dashboard
- Zero SQL query errors on execution
- Dashboard filterable by nationality, region, and demographic category
- Every step clearly documented in the GitHub README
- All files organized in a logical, well-structured folder hierarchy